Name of foam	Density (lb/ft³)		Density kg/m^3	Compressive Strength (lb/in²) at 25%	Compressive Strength (lb/in²) at 50%	Young's Modulus, E (psi)
Polyethylene Foam Sheet (Closed-Cell PE)	1.2		19.22	7	12	26
Polyethylene Foam Sheet (Closed-Cell PE)	1.7		27.23	7	14	28
Polyethylene Foam Sheet (Closed-Cell PE)	2.2		35.24	8	14.5	31
Polyethylene Foam Sheet (Closed-Cell PE)	4		64.07	16.5	20	53
Cross-linked Polyethylene Foam (XLPE)	2		32.04	8.2	17.1	34
Cross-linked Polyethylene Foam (XLPE)	3		48.06	10.5	19.8	41
Cross-linked Polyethylene Foam (XLPE)	4		64.07	21.2	31.8	74
Egg mass	0.06					
velocity	**6**					
maximum force	35					
Name of foam	**Density (lb/ft³)**			**Compressive Strength (lb/in²) at 25%**	**Compressive Strength (lb/in²) at 50%**	**Young's Modulus, E (psi)**
Polyethylene Foam Sheet (Closed-Cell PE)	1.2		19.22	7	12	26
Polyethylene Foam Sheet (Closed-Cell PE)	1.7		27.23	7	14	28
Polyethylene Foam Sheet (Closed-Cell PE)	2.2		35.24	8	14.5	31
Polyethylene Foam Sheet (Closed-Cell PE)	4		64.07	16.5	20	53
Cross-linked Polyethylene Foam (XLPE)	2		32.04	8.2	17.1	34
Cross-linked Polyethylene Foam (XLPE)	3		48.06	10.5	19.8	41
Cross-linked Polyethylene Foam (XLPE)	4		64.07	21.2	31.8	74
Decision matrix	Fast Loading Time (1-5)		Manufacturing Ease (1-5)	Lower Chances of Failures (1-5)	Total	
End Cap Opening	5		5	4	14	
Door Opening	3		2	2	7	
Decision matrix	Fast Loading Time (1-5)		Manufacturing Ease (1-5)	Lower Chances of Failures (1-5)	Cost	total
Foam Only	5		5	3	5	18
Suspension	2		2	1	2	7
Variation	Egg capacity (1-5)	Low Chance of breaking (1-5)	Cost (1-5)	Weight (1-5)	Manufacturing (1-5)	Total
XPLE 2 2 in	1	4	1	4	2	17
XPLE 2 3in	1	5	1	5	2	20
PE 2.2 3.5in	1	5	5	5	5	25.5
PE 2.2 2in EGG 3	5	1	5	1	5	25.5
PE 2.2 2in EGG 2	4	1	5	2	5	23.5
	1	5	3	5	5	23.5
Variation	Egg capacity		Cost	Weight (foam + eggs)	Manufacturing	Total
XPLE 2 2 in	3		$57.99	7.76 oz		
XPLE 2 3in	2		$77.99	5.79 oz		
PE 2.2 3.5in	2		$14.99	5.97 oz		
PE 2.2 2in EGG 3	9		$14.99	19.69 oz		
PE 2.2 2in EGG 2	6		$14.99	13.83 oz		
Version	**Foam Density (lb/ft³)**		**Foam-Only Weight (lb)**	**Foam-Only (oz)**	**Total (Foam + Eggs) (lb)**	**Total (oz)**
XPLE 2 – 3 eggs	2		0.107	1.71	0.1070 + 3 × 0.126 = 0.485	7.76 oz
XPLE 2 – 2 eggs	2		0.1101	1.76	0.1101 + 2 × 0.126 = 0.362	5.79 oz
PE 2.2 – 2 eggs	2.2		0.1212	1.94	0.1212 + 2 × 0.126 = 0.373	5.97 oz
PE 2.2 EGG 3 – 9 eggs	2.2		0.0972	1.56	0.0972 + 9 × 0.126 = 1.23	19.69 oz
PE 2.2 EGG 2 – 6 eggs	2.2		0.1075	1.72	0.1075 + 6 × 0.126 = 0.865	13.83 oz
Parts	weight	cost				
Egg		0				
total						

Poisson's Ratio, v	egg mass kg	Maximum force N	Velocity in m/s	Contact area	e allowed	Minimum distance of foam in cm
0.25	**0.06**	**35**	**6**	0.001	0.1952431168	31.60894311
0.25	**0.06**	**35**	**6**	0.001	0.1812971799	34.04040027
0.25	**0.06**	**35**	**6**	0.001	0.1637522915	37.68758602
0.25	**0.06**	**35**	**6**	0.001	0.0957796422	64.4336148
0.25–0.30	**0.06**	**35**	**6**	0.001	0.1493035599	41.33477176
0.25–0.30	**0.06**	**35**	**6**	0.001	0.1238127082	49.84487183
0.25–0.30	**0.06**	**35**	**6**	0.001	0.06859893292	89.96391501

Poisson's Ratio, v	egg mass kg	Maximum force N	Velocity in m/s	Contact area	e allowed	Minimum distance of foam in cm
0.25	**0.06**	**35**	**6**	0.001	0.1952431168	
0.25	**0.06**	**35**	**6**	0.001	0.1812971799	
0.25	**0.06**	**35**	**6**	0.001	0.1637522915	
0.25	**0.06**	**35**	**6**	0.001	0.0957796422	
0.25–0.30	**0.06**	**35**	**6**	0.001	0.1493035599	
0.25–0.30	**0.06**	**35**	**6**	0.001	0.1238127082	
0.25–0.30	**0.06**	**35**	**6**	0.001	0.06859893292	